Soybean

Glycine max — Pakistan's Emerging Protein and Oilseed Crop

A Complete Cultivation Guide for

Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan

Compiled for Farmer Guidance & AI Advisory Use

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1. Introduction

Soybean is a relatively newer but rapidly growing crop in Pakistan, valued both as an oilseed and as a high-protein feed and food source. Unlike wheat, rice, or cotton, soybean does not yet have a long tradition of cultivation across most of Pakistan, but government and research interest in the crop has grown significantly because of its potential to reduce the country's large edible oil and protein meal import bill. It is mainly grown as a summer (kharif) crop, fitting into rotations with wheat in parts of Punjab and Khyber Pakhtunkhwa. This guide explains how soybean is grown in Pakistan's conditions and the practices needed to establish it successfully.

2. Scientific Name

Soybean's scientific name is Glycine max, a member of the Fabaceae (legume) family, the same family as chickpea, making it a nitrogen-fixing crop that can improve soil fertility when properly inoculated with the correct Rhizobium (Bradyrhizobium japonicum) bacteria.

3. Importance in Pakistan

Soybean is important to Pakistan primarily because of the country's heavy reliance on imported edible oil and soybean meal for poultry and livestock feed; expanding domestic soybean production could meaningfully reduce this import dependence over time. It is being promoted as a rotational crop that can fit between wheat harvest and the next wheat sowing, making productive use of the summer window in parts of Punjab and Khyber Pakhtunkhwa without displacing other major crops. As a legume, it also contributes to improved soil fertility through nitrogen fixation, which can benefit the following wheat crop in the rotation, an added agronomic benefit beyond its direct market value.

4. Major Producing Provinces and Districts

- Punjab: Faisalabad, Sahiwal, Okara, and parts of the Pothohar region — the main areas where soybean cultivation and adaptation trials have expanded in recent years.
- Khyber Pakhtunkhwa: Peshawar valley, Mardan, and Swat valley — significant research and pilot cultivation given suitable summer temperatures.
- Sindh and Balochistan: Limited but growing interest, mainly through research trials and pilot programs, since Pakistan's overall soybean area remains modest compared to established oilseed crops.

5. Climate Requirements

Soybean grows best with warm temperatures between 20°C and 30°C during the growing season, with adequate but not excessive moisture, particularly during flowering and pod-filling. It is a day-length-sensitive crop, and choosing varieties adapted to Pakistan's latitude and day-length pattern is essential for proper flowering and pod set. Excessive heat above 35–38°C during flowering can reduce pod set, while waterlogging or drought stress at critical growth stages both significantly reduce yield.

6. Soil Requirements

Soybean grows best in well-drained, fertile loam to clay loam soils with a near-neutral pH (6.0–7.0), and it tolerates a range of soil textures better than many other legumes, provided drainage is adequate. Poorly drained or highly saline soils should be avoided since soybean is sensitive to both waterlogging and salinity, more so than crops like mustard.

7. Land Preparation

Land should be ploughed two to three times followed by planking to prepare a fine, well-leveled seedbed, since good soil-to-seed contact and adequate soil aeration are important for germination and for the nodulation process that supports nitrogen fixation. Fields should be properly leveled to avoid waterlogging in low spots, which can severely damage young soybean plants.

8. Recommended Varieties (Pakistan)

- Varieties tested and released by NARC's National Soybean Programme and provincial research institutes, selected for adaptation to Pakistan's day-length and temperature conditions.
- Williams-type and other introduced varieties adapted through local trials at agricultural universities and research stations in Punjab and Khyber Pakhtunkhwa.
- Because soybean cultivation is still developing in Pakistan compared to more established crops, farmers are strongly advised to obtain the latest variety recommendations directly from NARC, the University of Agriculture Faisalabad, or provincial extension departments before each season, since suitable varieties and seed sources are still expanding.

9. Seed Rate

Recommended seed rate for soybean is about 25–30 kg per acre, depending on seed size and germination percentage, sown to achieve a plant population sufficient for good canopy closure and weed suppression without excessive crowding.

10. Seed Treatment

Soybean seed should be treated with a fungicide such as thiram or carbendazim (2–3 g per kg seed) to protect against seedling diseases. Just as importantly, seed must be inoculated with the specific Bradyrhizobium japonicum culture required for soybean nodulation shortly before sowing, since this bacterium is generally not naturally present in Pakistani soils that have not grown soybean before; without proper inoculation, nitrogen fixation and yield are both significantly reduced.

11. Sowing Time (Province-wise)

- Punjab: Sown late June to July as a kharif crop, timed to follow wheat harvest and make use of monsoon rains.

- Khyber Pakhtunkhwa: Sown June–July in the plains and valleys where summer temperatures and rainfall patterns suit the crop.
- Sindh and Balochistan: Sowing windows are still being refined through ongoing research trials, generally targeting the summer monsoon period where soil moisture and temperature are favorable.

12. Sowing Methods

Soybean is typically drilled in rows about 45–60 cm apart with plant-to-plant spacing of around 5–7 cm, at a sowing depth of about 3–5 cm, using a seed drill for uniform spacing where available. Proper spacing supports good canopy development, which helps with natural weed suppression once the crop is established.

13. Fertilizer Recommendations (NPK)

Because soybean fixes a significant share of its own nitrogen needs when properly inoculated, a general recommendation is a modest starter dose of 20–25 kg Nitrogen, along with 60 kg Phosphorus (P_2O_5) and 30–40 kg Potash (K_2O) per acre, applied fully at sowing. Phosphorus is particularly important for good root and nodule development, so soil testing to confirm adequate phosphorus availability is recommended before finalizing the fertilizer plan.

14. Irrigation Schedule

Since soybean is grown mainly as a monsoon-season crop, irrigation needs depend heavily on rainfall; where rainfall is insufficient, irrigation every 10–15 days is generally recommended, with special attention to the flowering and pod-filling stages, which are the most sensitive to both water stress and waterlogging. Fields must have good drainage to remove excess water quickly after heavy monsoon rains, since standing water for even a short period can seriously damage the crop.

15. Weed Management

Young soybean plants are relatively poor competitors against weeds until the canopy closes, so weed control is critical in the first four to six weeks after sowing, typically through one to two hand weedings or hoeings. Because sowing coincides with the monsoon season when weed pressure is naturally high, timely weeding is especially important for achieving good yield in Pakistani conditions.

16. Insect Pests

Common insect pests reported on soybean in Pakistan's trial and pilot production areas include whitefly, jassids, stem fly, and pod borer caterpillars, some of which are shared with other summer legume and cotton crops grown in the same regions. Regular field monitoring and need-based insecticide application, along with crop rotation to avoid buildup of shared pests with cotton or other legumes, are recommended management practices.

17. Diseases

Soybean can be affected by root rot and stem rot under waterlogged conditions, along with some viral diseases transmitted by whitefly, similar to other legume and vegetable crops grown in the same season. Because soybean is still a developing crop in Pakistan, disease pressure and specific local strains are still being studied by research institutes, so farmers are advised to follow the latest disease advisories from NARC and provincial agriculture departments and to prioritize good field drainage as a general precaution.

18. Deficiency Symptoms

Nitrogen deficiency, when nodulation has failed due to poor or absent inoculation, shows as overall pale yellow-green leaves and stunted growth. Phosphorus deficiency causes poor root and nodule development and delayed maturity. Potassium deficiency appears as yellowing and scorching along leaf margins. Iron deficiency chlorosis (yellowing between leaf veins on young leaves) can occur on the calcareous, alkaline soils common in parts of Punjab, a known challenge for soybean production in similar soil types elsewhere in the world.

19. Harvesting

Soybean is ready for harvest when most pods turn brown and dry and the leaves yellow and drop, typically 90–120 days after sowing depending on variety. Plants are cut by hand or harvested with combines where mechanization is available, then dried further if needed before threshing, with care taken to harvest promptly once mature since delayed harvest in humid conditions can cause pod shattering and seed quality loss.

20. Storage

Threshed soybean should be dried to a safe moisture level (below about 12%) before storage to prevent mold growth and preserve both seed viability and oil quality. Seed should be stored in clean, dry, well-ventilated conditions, protected from moisture, insect pests, and rodents, since soybean's oil and protein content make it attractive to storage pests if not properly protected.

21. Expected Yield in Pakistan

Because soybean cultivation in Pakistan is still at a relatively early and expanding stage compared to more established crops, yields vary considerably between research trials and farmer fields, but well-managed plots with proper inoculation, adapted varieties, and adequate moisture have shown yields broadly comparable to other summer legume crops in similar agro-climates. Yield potential is expected to improve further as locally adapted varieties, inoculant availability, and farmer experience with the crop continue to develop.

22. Government Recommendations

NARC's National Soybean Programme, along with the University of Agriculture Faisalabad and provincial agriculture departments, is actively working to identify suitable varieties, refine agronomic packages, and expand extension support for soybean as part of Pakistan's broader strategy to reduce edible oil and protein meal imports. Farmers interested in growing soybean are encouraged to connect with these research institutes for the latest variety and inoculant recommendations, since this crop's package of practices is still being refined for Pakistani conditions compared to long-established crops.

23. Modern Farming Practices

Because soybean is a newer crop in Pakistan, its expansion has been closely linked with modern practices from the start, including proper Rhizobium inoculation, soil testing before fertilizer application, and row sowing with seed drills for uniform stands. Some pilot programs are also testing soybean-wheat rotations to demonstrate the soil fertility and system-level benefits of including a nitrogen-fixing summer crop ahead of wheat.

24. Precision Agriculture

Precision approaches being piloted for soybean in Pakistan include soil testing to guide phosphorus application and confirm suitability for inoculation, and monitoring of rainfall and soil moisture to help time supplemental irrigation during flowering and pod-fill. As soybean area expands, research institutes are expected to extend the same satellite and mobile-advisory tools already used for other major crops to help guide sowing dates and pest alerts for soybean growers.

25. Sustainable Farming

Soybean's nitrogen-fixing ability makes it a genuinely sustainable addition to Pakistan's cropping systems, particularly in rotation with wheat, since it can reduce the following crop's nitrogen fertilizer requirement while diversifying farm income and reducing reliance on imported protein meal and edible oil. Encouraging proper inoculation instead of relying on synthetic nitrogen, promoting locally adapted varieties suited to Pakistan's day-length and climate, and integrating the crop thoughtfully into existing wheat-based rotations are the key sustainability priorities as the crop's cultivation expands.

26. Regional Notes

Soybean remains an emerging rather than fully established crop in Pakistan, with the most promising results so far seen in parts of Punjab and Khyber Pakhtunkhwa where summer temperatures, monsoon rainfall, and existing wheat-based rotations align well with the crop's requirements. Farmers considering soybean for the first time are strongly encouraged to start with recommended varieties, proper Rhizobium inoculation, and close contact with NARC or university extension services, since the crop's full potential in Pakistan is still being demonstrated and refined through ongoing research and pilot programs.